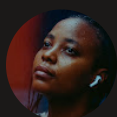




Creating a Private Subnet



Lenah Mbatha

VPC / Subnets / Create subnet

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block
 256 IPs



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a virtual private cloud which provides an isolated virtual network in the AWS cloud. It is useful because it's where users can launch their AWS Resources

How I used Amazon VPC in this project

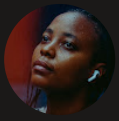
In today's project, I used Amazon VPC to create a Private subnet, a private network ACL

One thing I didn't expect in this project was...

One thing I didn't expect in this project is the simplicity of creating a private network ACL

This project took me...

This project took me 30 minutes



Private vs Public Subnets

The difference between public and private subnets is that for public subnets, the resources inside them are accessible by the public eg a static website while a private subnet, the resources inside them aren't accessed by the public eg company's database

Having private subnets are useful because they keep resources that do not want to be accessed public

My private and public subnets cannot have the same subnets due to an overlap. when private and public subnets are the same, there would be conflicts

VPC / Subnets / Create subnet

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Nextwork Private Subnet
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Europe (Stockholm) / eu-north-1b

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.1.0/24 256 IPs

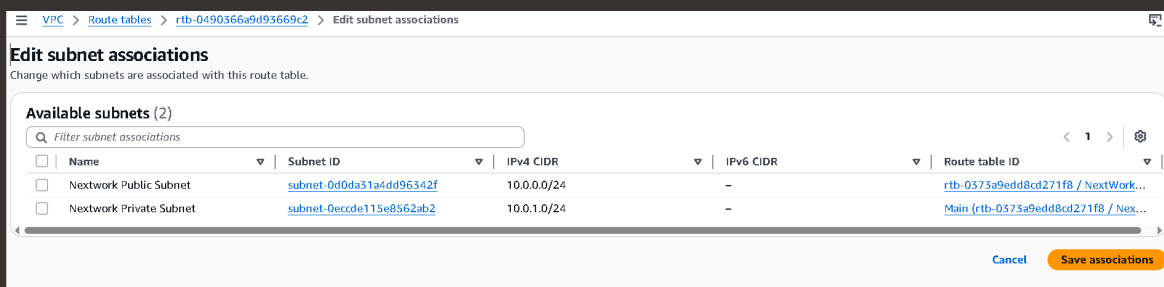


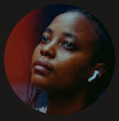
A dedicated route table

By default, my private subnet is associated with my VPC's route table

I had to set up a new route table because NextWork route table has a route to an internet gateway, and having that route would make your subnet public

My private subnet's dedicated route table only has one inbound and one outbound rule that allows local traffic within the VPC only (using the CIDR block of the VPC, e.g., 10.0.0.0/16, with a target of "local"). There is no route to an internet gateway, which ensures that the private subnet remains isolated from the public internet. All traffic destined for or originating from outside the VPC is implicitly denied, keeping private resources secure and inaccessible to external networks.



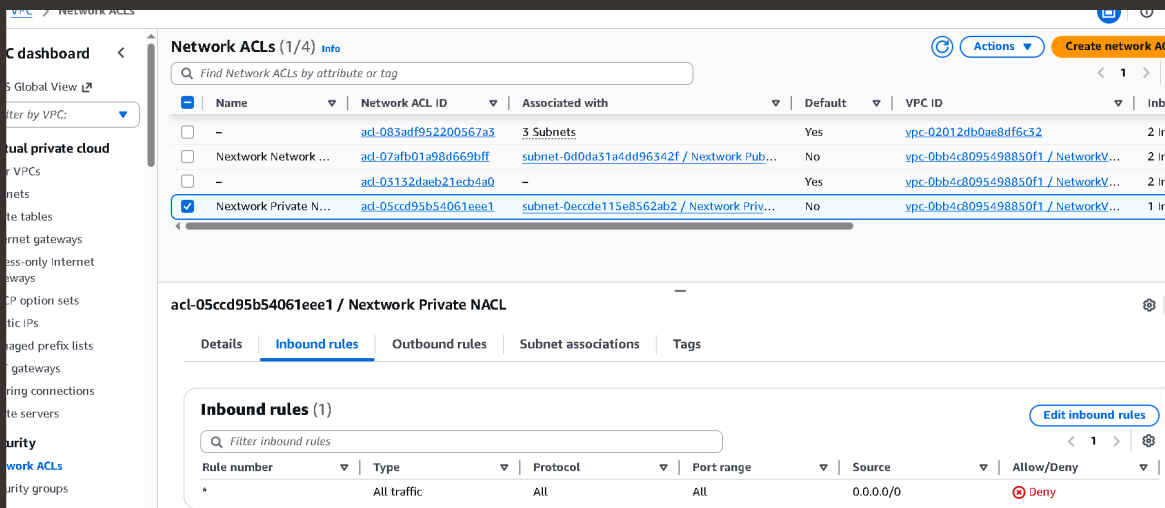


A new network ACL

By default, my private subnet is associated with the default Network ACL

I set up a dedicated network ACL for my private subnet because the default network ACL allows all traffic to go in my subnet and out of my subnet and this will make an attacker access private resources in my private subnet

My new network ACL has two simple rules - Inbound rules these are instructions that govern the traffic coming into your resources while outbound rules are rules that govern the traffic that comes out of your resources





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